

Reproduction: Topic 1

Econ E673: Micro-Econometrics, Fall 2026

Topic I. Dehejia & Wahba (1999, *JASA*), reanalyzing LaLonde (1986, *AER*)

Reproduce: the treatment effects estimated in the paper.

1. The experimental benchmark: the treatment-control difference in 1978 earnings within the NSW sample (`nsw_dw.dta`). The published value is \$1,794.
2. Dehejia and Wahba's propensity-score ATT for the NSW treated against the CPS comparison group (the treated units from `nsw_dw.dta` plus `cps_controls.dta`), set against the matching estimates in their results table.

The two NSW samples. There are two experimental samples in circulation and they carry different right answers. LaLonde's male sample is 297 treated and 425 controls. Dehejia and Wahba restricted it to men with two years of pre-treatment earnings, which leaves 185 treated and 260 controls, and that subsample is `nsw_dw.dta`. The experimental benchmark is about \$886 in LaLonde's sample and \$1,794 in Dehejia and Wahba's. Check which file you have opened before concluding that your code is wrong.

Data: the paper's own files, posted by Rajeev Dehejia and mirrored on Canvas: `nsw_dw.dta` and `cps_controls.dta`, plus `nsw.dta` (LaLonde's original experimental sample, needed for the second half below) and `psid_controls.dta` if you want the PSID comparison. There is no author code for this topic. I have posted a sample do-file to start you off, but check every line of it, and edit it to match your own understanding of the algorithm.

The archive. All the data comes from Dehejia's NSW archive, <https://users.nber.org/~rdehejia/nswdata2.html>, which also posts the narrower comparison groups if you want more samples to move across: PSID2 (253), PSID3 (128), CPS2 (2,369), CPS3 (429). The data are distributed for non-commercial use under CC BY-NC, and the archive asks that you cite three papers: Dehejia and Wahba (1999, *JASA*), Dehejia and Wahba (2002, *ReStat*), and LaLonde (1986, *AER*). Cite all three.

The second half: audit it the way Smith and Todd did

Smith and Todd (2005) is the Week 3 discussant paper. Your half page runs their three checks on your own estimates.

1. **Overlap.** Plot the two $\hat{e}$ densities. Enforce common support, report how many comparison units you dropped, and say what population your ATT now describes.
2. **Balance.** Within propensity-score strata, compare covariate means across treated and controls. Report balance before and after your adjustment, and say which covariates required refitting the score.

3. **Sample selection.** You have two experimental files: `nsw.dta` (LaLonde's sample, 297 treated / 425 controls) and `nsw_dw.dta` (Dehejia and Wahba's subsample, 185 / 260). From the two papers, explain why Dehejia and Wahba imposed the two-years-of-earnings restriction, and say what it does to the population the estimate describes. Then re-estimate your matched ATT on *LaLonde's* sample, where `re74` does not exist and cannot enter the score, and compare the three numbers: benchmark, DW-sample ATT, LaLonde-sample ATT. What you find is the discussant's argument, in your own output.

State your matching choices: matching method, number of matches, whether matching is with replacement, and your common-support rule (including any caliper or trimming rule).

What to submit

One file per group, per exercise:

- **The reproduced object.** The table or figure, next to the published version, so the comparison is visible.
- **The audit.** What you checked, what you found, and what stopped you where something did.
- **Your code,** in whatever language you like. It should run start to finish on a clean machine given the package's data.

Name it `E673_repro_TopicN_LastnameLastnameLastname`.